

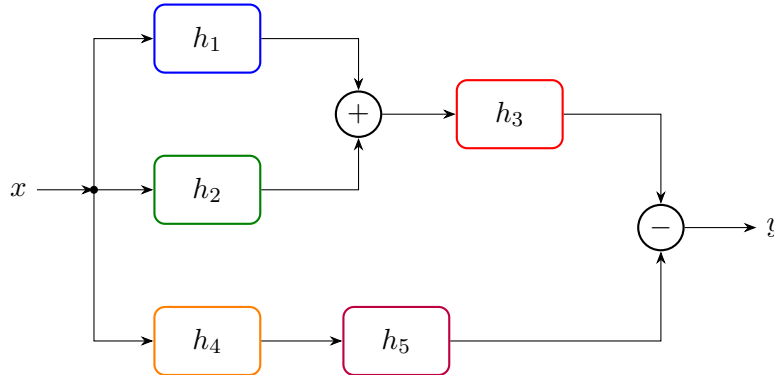
CSE220
Online on Convolution

Subsections: C1, C2

Total Marks: 10

Total Time: 30 Minutes

This online reuses your `DiscreteSignal` and `LTISystem` classes from the offline. You will use them to calculate the output of a system built from four sub-systems combined as shown below.



Tasks

- Use your completed `DiscreteSignal` and `LTISystem` classes from the offline.
- Determine and plot the output of the combined system by applying h_1, h_2, h_3, h_4, h_5 block by block, exactly as shown in the diagram. Use only `DiscreteSignal`/`LTISystem` methods.
- Determine `h_combined`, the single impulse response of an LTI system equivalent to the entire diagram, again using only `DiscreteSignal`/`LTISystem` operations. Determine and plot the output of applying `h_combined` directly to x .
- Verify that both outputs are the same signal by computing the maximum absolute difference between them. Formula: $\max_n |y_{\text{block}}[n] - y_{\text{combined}}[n]|$.

Marks Distribution

- Block-by-block output (y_{block}): 4
- Correct `h_combined` and output via combined system (y_{combined}): 4
- Verification that $y_{\text{block}} = y_{\text{combined}}$: 2

Submission

Rename your python file as 2305xxx.py and submit in the moodle.